PRODUCT USER MANUAL

Arctic Sea Ice Age

SEAICE_ARC_PHY_SIAGE_L4_MY_011_025
Issue: 1.0

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RECORD TABLE

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GLOSSARY AND ABBREVIATIONS

TAC	Thematic Assembly Center
ECMWF	European Centre for Medium Range Weather forecast
NRT	Near Real Time
MY	Multi Year
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus Marine Environment Monitoring Service
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
NRT-ANALYSIS:	An hindcast done with data assimilation for a period very close to present. The quality and the number of available observations it's not optimal (less observations and/or less accurate than in a best-analysis).
BEST-ANALYSIS:	An hindcast done with data assimilation for a period close to present. The quality and the number of available observations it's the best available (more observations and/or more accurate than in a NRT-analysis).
MYI	Multi-Year Ice
FYI	First-Year Ice
SIAge	Sea Ice Age
GCOS	Global Climate Observing System
EASE	Equal-Area Scalable Earth
OSI SAF	Ocean and Sea Ice Satellite Application Facilities

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download. Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

1) INTRODUCTION

a) Summary

This document describes the multi-year product Arctic Sea Ice Age (SIAge) produced by the Nansen Environmental and Remote Sensing Centre in Bergen, Norway. The dataset temporal coverage begins on 15/09/1995 and is updated annually, extending up to the year preceding the current one (N-1). The dataset is based on a novel Lagrangian advection scheme applied to a triangular mesh, which conserves sea ice age fractions and reduces numerical diffusion compared to the previous Eulerian approach. SIAge is derived from satellite observations of sea ice concentration and drift, and represents fractional age classes per grid cell. The record captures the spatial and temporal evolution of first- to sixth-year ice, including uncertainty estimates that account for both sea ice concentration and drift uncertainties.

Validation of SIAge using ice drifting buoys indicates good consistency, as the product does not underestimate the maximum age of the buoys in 98.3% of cases, with most discrepancies occurring near the ice edge.

The SIAge dataset improves the observational basis for Arctic monitoring and contributes to the Global Climate Observing System (GCOS) Essential Climate Variables.

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	Arctic Sea Ice Age
Geographical coverage	Arctic: 180°W-180°E ; 60°N – 90°N
Variables	• Average sea ice age • Concentration of first-year ice • Concentration of second-year ice • Concentration of third-year ice • Concentration of fourth-year ice • Concentration of fifth-year ice • Concentration of sixth-year ice • Concentration of seventh-year ice
Product Type	Multi-year, satellite observations
Available time series	15/09/1995 – -1M
Temporal resolution	Daily
Update Frequency	Monthly
Target delivery time	First Monday of the month at 12:00
Horizontal resolution	65 km
Number of vertical levels	1
Format	NetCDF4

b) Production System Description

We compute sea ice age from satellite-derived sea ice concentration and drift in the following way. A pan-Arctic field of concentration is taken from satellite observations at the end of the melt season, when the ice extent is minimal, and all ice is assumed to be multi-year ice (MYI, C_{MY}) (Fig. 1, Step 1). The C_{MY} field is repetitively advected (morphed) using daily satellite-observed sea ice drift fields (Fig. 1, Step 2). Changes of concentration due to ice deformation or melting are accounted for during the advection process. Some time after the initialization of the MYI field advection (e.g. on 1 Jan 1992, as shown in Figure 1), the advected field represents concentration of MYI which is lower than the total observed concentration and is denoted as C_{A0} (i.e., advected for less than one year). The difference between the total and advected fields yields the concentration of the first-year ice (Fig. 1, Step 3):

$$C_{1Y} = C_{TOT} - C_{A0}.$$

Hereafter, we define the first-year ice (C_{1Y}) as ice formed during the ongoing freezing season and that has not yet experienced melting.

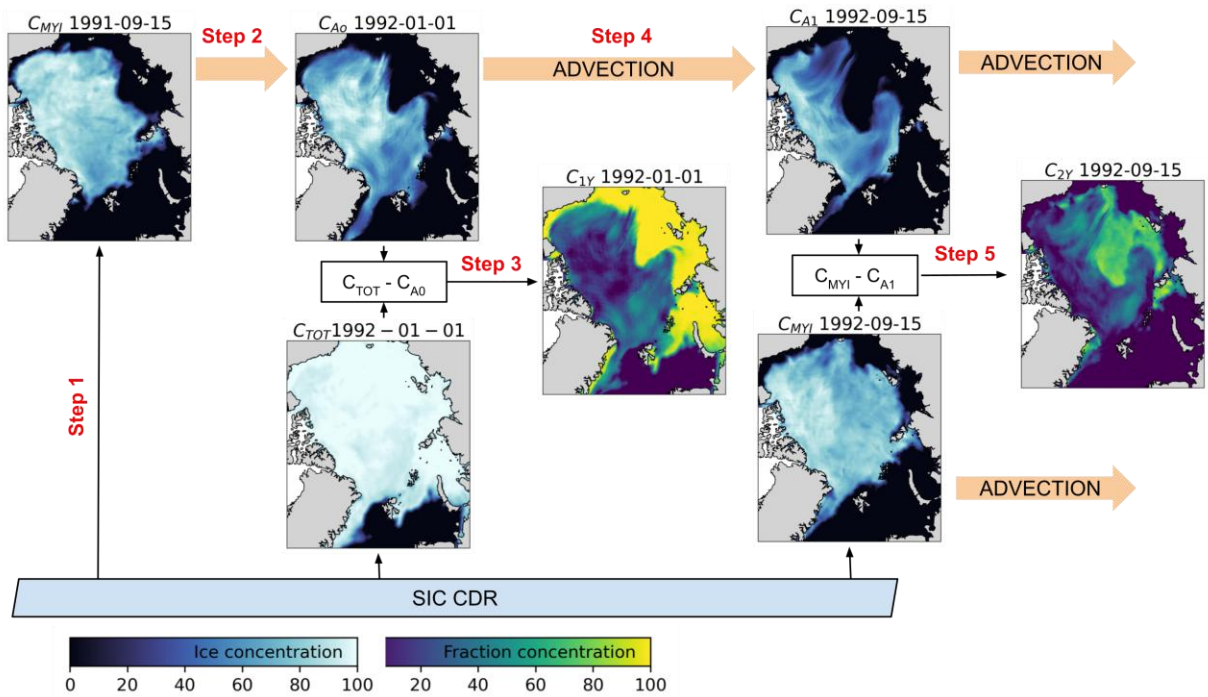

Figure 1. *A scheme for sea ice age fraction computation within one year. The fields of MYI and TOT concentration originate from SIC CDR (Ice concentration colour map). These fields are advected using sea ice drift from the SID CDR (advection is shown by thick orange arrows). The advected fields of MYI are labelled as "C_{Ai} YYYY-MM-DD", where i denotes the number of years of advection and YYYY-MM-DD denotes the date of the advected field. MYI and advected fields are used to compute sea ice age fractions (Fraction concentration colourmap), labeled as "C_{jY} YYYY-MM-DD", where j denotes the age of the ice fraction, i.e. C_{1Y} stand for the first-year ice and C_{2Y} stands for the second-year ice.*

One year after the initialisation, the advected field (Fig. 1, Step 4) represents the concentration of sea ice which is at least two years old and is denoted C_{A1} (advected for one year). At that time, the total observed concentration again reaches a minimum, representing the concentration of multi-year ice. C_{MY} has higher values than the advected C_{A1} as it also contains a fraction of the second-year ice (Fig. 1, Step 5):

$C_{2Y} = C_{MY} - C_{A1}$.

Hereafter, we define the second-year ice (C_{2Y}) as ice formed during the previous freezing season and that has survived one melting season. It should be noted that according to the nomenclature of the World Meteorological Organisation (Sea Ice Nomenclature, WMO-259), the first-year ice (FYI) that survives the summer minimum is called "residual ice", and it turns into second-year ice only on 1 January of the coming winter. Nevertheless, hereafter, all survived FYI turns into the second year after 15 September.

Both C_{A1} and the current C_{MY} are advected further using the sea ice drift and after one more year they become C_{A2} and C_{A1}, i.e. ice fractions advected for two years and for one year. Since C_{A1} contains C_{A2} and, similar to the previous year, the new C_{MY} contains C_{A1} we can compute fractions of the second- and third-year ice:

$C_{2Y} = C_{MY} - C_{A1}$

$C_{3Y} = C_{A1} - C_{A2}$

This workflow is repeated, and the ice age fraction can be computed using a generic formula:

$C_{NY} = C_{A(N-2)} - C_{A(N-1)}$,

where N is an integer number indicating the age of sea ice fraction.

In this algorithm, we use a Lagrangian scheme of advection of MYI ice, where nodes of a triangular mesh are advected using the ice drift vectors. Triangular elements in this mesh are advected without diffusion unless a remeshing occurs. A detailed description of the input data, advection, remeshing and validation is provided in Korosov et al., 2026.

The algorithm for mesh advection was initialized on 1 January 1991 and advected the mesh using SID CDR until 2020, and using the NRT SID product until. The MYI concentration field was computed for 15 September for years 1991 – 2024. Each MYI concentration field is advected for 2200 days (6 years). Thus, for each day after 15 September 1995, we obtained 6 fields of C_{A1}–C_{A5}. For the spin-up period before 15 September 1995, we had fewer fields with advected MYI. Since the same meshes and mappings were used for advection of MYI concentration originating from different years, for each day, the C_{A1}–C_{A5} fields are located on the same mesh, which facilitates computation of sea ice age fractions, see Eqs. above. For dates between the minimum ice extent, the first-year ice concentration was also computed.

The sea ice age is then computed using a weighted average as suggested in our previous algorithm:

$A = \Sigma_i A_i C_{iY} / \Sigma_i C_{iY}$

where $A_i \in [1,2,3,4,5,6,7]$ is an integer year, C_{iY} is the fraction of ice of corresponding age (C_{1Y} corresponds to FYI).

Values of the ice age fractions, as well as the average ice age, were linearly interpolated on a regular grid in the EASE2 projection with a resolution of 65 km and exported to netCDF files.

c) Dataset details

Product Lines	SEAICE_ARC_PHY_SIAGE_L4_MY_011_025	
Geographical coverage	Arctic	
Datasets: cmems_obs-si_arc_phy-siage_my		
Product Type	Observations	
Update frequency	Monthly	
Available time series	1995 – -1M	
Target delivery time	First Monday of the month at 12:00	
Temporal resolution	Daily mean	

Dataset : cmems_obs-si_arc_phy-siage_my contains the 2D daily mean fields: average sea ice age, concentration of first-year ice, concentration of second-year ice, concentration of third-year ice, concentration of fourth-year ice, concentration of fifth-year ice, concentration of sixth-year ice
siage [years] average sea ice age age_of_sea_ice
conc_1y [%] concentration of first-year ice fisrt_year_sea_ice_area_fraction
conc_2y [%] concentration of second-year ice second_year_sea_ice_area_fraction
conc_3y [%] concentration of third-year ice third_year_sea_ice_area_fraction
conc_4y [%] concentration of fourth-year ice fourth_year_sea_ice_area_fraction
conc_5y [%] concentration of fifth-year ice fifth_year_sea_ice_area_fraction
conc_6y [%] concentration of sixth-year ice sixth_year_sea_ice_area_fraction
conc_7y [%] concentration of seventh-year ice seventh_year_sea_ice_area_fraction

Figures 2 and 3 show an example content of the file with sea ice age information: weighted average sea ice age (years) and concentrations of ice age fractions (%).

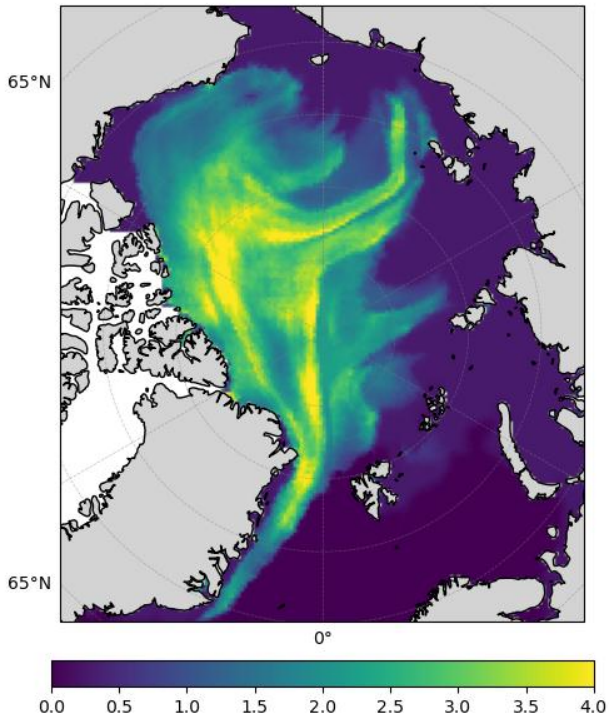

Figure 2. Example content of sea ice age file for 1 January 2000. The file contains the weighted average sea ice age shown in this figure and the concentrations of ice age fractions (see Fig. 3),

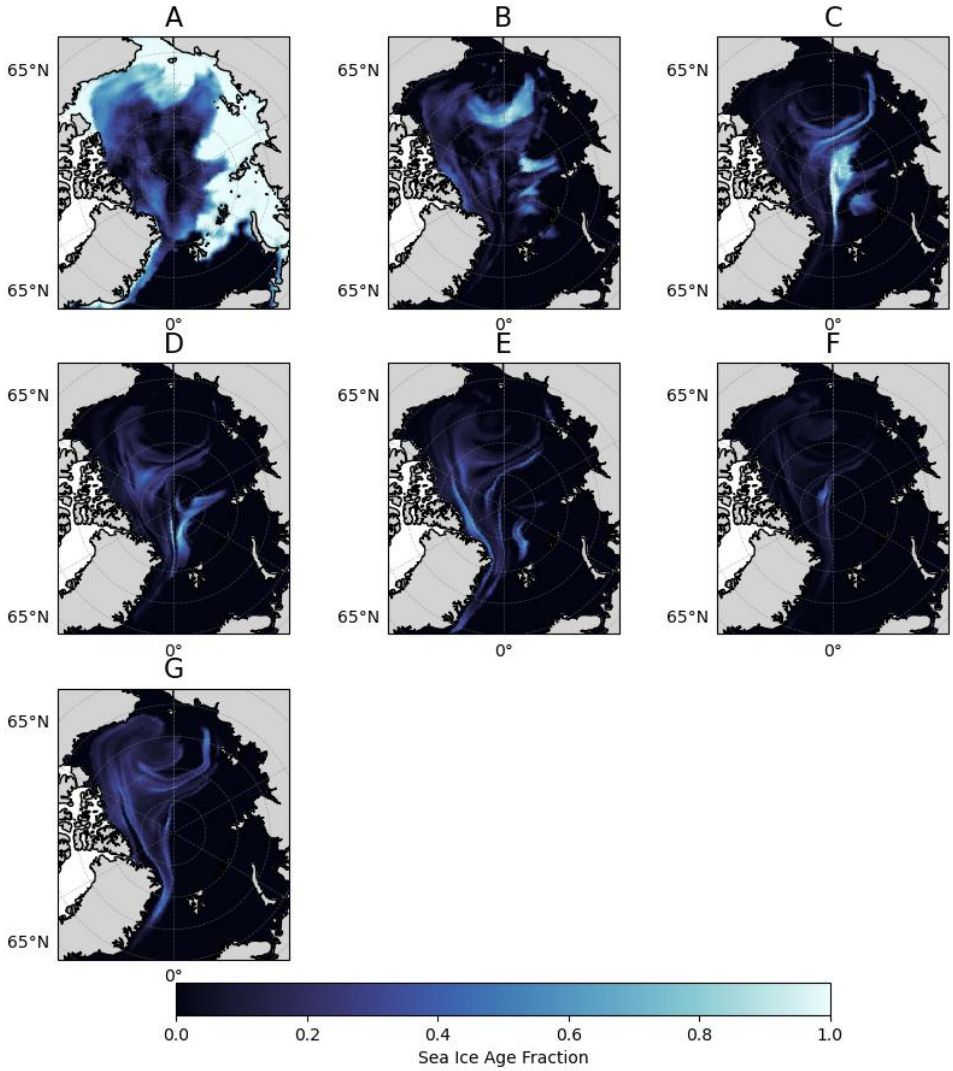

Figure 3. Example content of sea ice age file for 1 January 2000. The file contains the weighted average of sea ice age (see Fig. 2) and the concentrations of seven ice age fractions, as shown in the figure. A - fraction of the first-year ice. B - fraction of the second-year ice. C - fraction of the third-year ice. D - fraction of the fourth-year ice. E - fraction of the fifth-year ice. F - fraction of the sixth-year ice. G - fraction of the seventh-year ice.

d) Input satellite data

Sea Ice Drift data from EUMETSAT OSI SAF

For the period from 1991 through 2020 we use OSI-455, the EUMETSAT Ocean and Sea Ice Satellite Application Facilities (OSI SAF) global low-resolution sea ice drift climate data record (CDR) (OSISAF, 2022c; Lavergne and Down, 2023). For 2021–2024, we utilise the Global Low Resolution Sea Ice Drift product, OSI-405-c, which is a near-real time (NRT) product (OSISAF, 2007; Lavergne et al., 2010). Both products utilise the same satellite sensors, SSM/I (>= F10) and SSMIS (CMSAF FCDR), AMSR-E (NSIDC) and AMSR2 (JAXA). The NRT product also utilises C-band scatterometer data from the ASCAT missions. Both products employ the same sea ice motion-tracking methodology, continuous maximum cross-correlation (CMCC, Lavergne et al. (2010)), which involves a fractional-pixel pattern matching of the brightness temperatures.

These brightness temperatures are pre-processed with Laplacian filters, and the vectors are then post-processed with correction schemes. The product configurations differ between the CDR and NRT products. The CDR utilises an EASE-2 75 km grid, whereas the NRT product is based on a Polar Stereographic grid of 62.5 km. There is also a mismatch between the periods over which the motion vectors are retrieved, with the CDR using a period of 24 hours and the NRT product using a period of 48 hours. The final difference is that, due to the difficulty of retrieving vectors directly from the satellite data in the summers during the pre-AMSR-E period, a summer gap-filling method is used for the CDR. The summer motion vectors of the CDR are retrieved from an implementation of the free-drift model (Thorndike and Colony, 1982; Thomas, 1999; Brunette et al., 2022), using the ERA5 wind reanalysis and trained on the motion vectors from running CMCC on AMSR-E (2002-2011) and AMSR2 (2012-) brightness temperatures.

Sea Ice Concentration from EUMETSAT OSI SAF

The global sea ice concentration climate data record version 3 from SMMR/SSMI/SSMIS data (Lavergne et al., 2019; OSISAF, 2022b) was used for dates up to and including 2020, and the accompanying interim climate data record (ICDR) version 3 (OSISAF, 2022a), based on SSMIS data, for dates 2021 – 2024. These products are both on a 25 km EASE2 grid (Brodzik and Knowles, 2011). These products are retrieved using the SICCI3LF algorithm based on the 19 GHz and 37 GHz imagery. Information about the land/water mask was obtained from the status flag of the OSI SAF SIC product.

e) Product use and impact

The LM-SIAge dataset improves the observational basis for Arctic monitoring and contributes to the Global Climate Observing System (GCOS) Essential Climate Variables. The rapid Arctic warming in the early 2000s triggered a tremendous loss of multi-year sea ice. Since then, sea ice age has been largely overlooked, despite its value in evaluating climate model skill in capturing sea ice dynamics. An ice-tracking algorithm can be used to diagnose sea ice age from daily concentration and drift data in a climate model and to compare it to satellite-derived estimates using the same method. The derived sea ice age retains the spatial pattern and long-term trends of the reported model age, but with less than half the bias. Changes in sea-ice

age highlight shifts in sea-ice dynamics more clearly than thickness or volume. This shows that using a common algorithm enables both a fair model validation and improves insight into sea-ice dynamics (Davy et al., 2025).

3) FILE FORMAT

The products are stored using the NetCDF format.
To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF
To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

An exhaustive description of the files nomenclature when downloaded through the web portal is available here : Files Nomenclature and Format

More information on product and dataset nomenclatures is available here Products and Datasets nomenclature

5) FILE SIZE

DATASET NAME	NAME OF FILE	SIZE [MB]
cmems_obs-si_arc_phy-siage_my	cmems_obs-si_arc_phy-siage_my_YYYYMMDD.nc	1

6) STRUCTURE OF FILES

Examples of the header of output NetCDF files are inserted in **annex**, for each dataset.

7) REFERENCES

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- https://documentation.marine.copernicus.eu/QUID/CMEMS-SEAICE-QUID-011-025.pdf
-
-
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- Brunette, C., Tremblay, L. B., and Newton, R.: A new state-dependent parameterization for the free drift of sea ice, The Cryosphere, 16, 533–557, https://doi.org/10.5194/tc-16-533-2022, 2022.
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- Thorndike, A. S. and Colony, R.: Sea ice motion in response to geostrophic winds, Journal of Geophysical Research, 87, 5845–5852, https://doi.org/10.1029/JC087iC08p05845, 1982.
- Thomas, D.: The quality of sea ice velocity estimates, Journal of Geophysical Research, 104, 13627–13652, https://doi.org/10.1029/1999jc900086, 1999.

8) ANNEX

a) Structure of files

For dataset cmems_obs-si_arc_phy-siage_my_P1D-m

```
netcdf nersc_siage_arctic65_20250701 {
dimensions:
	y = 144 ;
	x = 144 ;
	time = UNLIMITED ; // (1 currently)
	nv = 2 ;
variables:
	byte Lambert_Azimuthal_Equal_Area ;
		Lambert_Azimuthal_Equal_Area:grid_mapping_name =
"lambert_azimuthal_equal_area" ;
		Lambert_Azimuthal_Equal_Area:longitude_of_projection_origin = 0. ;
		Lambert_Azimuthal_Equal_Area:latitude_of_projection_origin = 90. ;
		Lambert_Azimuthal_Equal_Area:false_easting = 0. ;
		Lambert_Azimuthal_Equal_Area:false_northing = 0. ;
		Lambert_Azimuthal_Equal_Area:semi_major_axis = 6378137. ;
		Lambert_Azimuthal_Equal_Area:inverse_flattening = 298.257223563 ;
		Lambert_Azimuthal_Equal_Area:proj4_string = "+proj=laea +lat_0=90 +lon_0=0
+x_0=0 +y_0=0 +datum=WGS84 +units=m +no_defs +type=crs" ;
	double y(y) ;
		y:standard_name = "projection_y_coordinate" ;
		y:units = "m" ;
		y:axis = "Y" ;
	double x(x) ;
		x:standard_name = "projection_x_coordinate" ;
		x:units = "m" ;
		x:axis = "X" ;
	double time(time) ;
		time:units = "seconds since 1978-01-01 00:00:00" ;
		time:long_name = "reference time of product" ;
		time:standard_name = "time" ;
		time:axis = "T" ;
		time:calendar = "standard" ;
		time:bounds = "time_bnds" ;
	double time_bnds(time, nv) ;
	float siage(time, y, x) ;
		siage:standard_name = "age_of_sea_ice" ;
		siage:long_name = "Weighted Average of Sea Ice Age" ;
		siage:name = "siage" ;
		siage:comment = "The weighted average is computed over all available fractions." ;
		siage:units = "years" ;
		siage:valid_min = 0.f ;
		siage:valid_max = 20.f ;
		siage:ancillary_variables = "sea_ice_age_uncertainty status_flag" ;
		siage:coverage_content_type = "modelResult" ;
		siage:grid_mapping = "Lambert_Azimuthal_Equal_Area" ;
```

```
        byte status_flag(time, y, x) ;
                status_flag:standard_name = "age_of_sea_ice status_flag" ;
                status_flag:long_name = "status flag array for sea ice age" ;
                status_flag:valid_min = 1b ;
                status_flag:valid_max = 3b ;
                status_flag:grid_mapping = "Lambert_Azimuthal_Equal_Area" ;
                status_flag:flag_masks = 1b, 2b, 3b ;
                status_flag:flag_meanings = "nominal land invalid" ;
                string status_flag:flag_descriptions = "flag = 1: Nominal retrieval by the SIA
algorithm", "flag = 2: Position is over land", "flag = 3: Pixel is invalid" ;
                status_flag:coverage_content_type = "qualityInformation" ;
        float longitude(y, x) ;
                longitude:standard_name = "longitude" ;
                longitude:long_name = "longitude" ;
                longitude:units = "degrees_east" ;
                longitude:grid_mapping = "Lambert_Azimuthal_Equal_Area" ;
        float latitude(y, x) ;
                latitude:standard_name = "latitude" ;
                latitude:long_name = "latitude" ;
                latitude:units = "degrees_north" ;
                latitude:grid_mapping = "Lambert_Azimuthal_Equal_Area" ;
        short conc_1yi(time, y, x) ;
                conc_1yi:_FillValue = -32768s ;
                conc_1yi:long_name = "Concentration of First Year Sea Ice" ;
                conc_1yi:name = "conc_1yi" ;
                conc_1yi:units = "1" ;
                conc_1yi:valid_min = 0s ;
                conc_1yi:valid_max = 1100s ;
                conc_1yi:ancillary_variables = "conc_1yi_uncertainty status_flag" ;
                conc_1yi:coverage_content_type = "modelResult" ;
                conc_1yi:standard_name = "sea_ice_area_fraction" ;
                conc_1yi:scale_factor = 0.001f ;
                conc_1yi:add_offset = 0.f ;
                conc_1yi:grid_mapping = "Lambert_Azimuthal_Equal_Area" ;
        short conc_2yi(time, y, x) ;
                conc_2yi:_FillValue = -32768s ;
                conc_2yi:long_name = "Concentration of Second Year Sea Ice" ;
                conc_2yi:name = "conc_2yi" ;
                conc_2yi:units = "1" ;
                conc_2yi:valid_min = 0s ;
                conc_2yi:valid_max = 1100s ;
                conc_2yi:ancillary_variables = "conc_2yi_uncertainty status_flag" ;
                conc_2yi:coverage_content_type = "modelResult" ;
                conc_2yi:standard_name = "sea_ice_area_fraction" ;
                conc_2yi:scale_factor = 0.001f ;
                conc_2yi:add_offset = 0.f ;
                conc_2yi:grid_mapping = "Lambert_Azimuthal_Equal_Area" ;
        short conc_3yi(time, y, x) ;
                conc_3yi:_FillValue = -32768s ;
                conc_3yi:long_name = "Concentration of Third Year Sea Ice" ;
                conc_3yi:name = "conc_3yi" ;
                conc_3yi:units = "1" ;
                conc_3yi:valid_min = 0s ;
                conc_3yi:valid_max = 1100s ;
```

```
        conc_3yi:ancillary_variables = "conc_3yi_uncertainty status_flag" ;
        conc_3yi:coverage_content_type = "modelResult" ;
        conc_3yi:standard_name = "sea_ice_area_fraction" ;
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        conc_3yi:add_offset = 0.f ;
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short conc_4yi(time, y, x) ;
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        conc_4yi:ancillary_variables = "conc_4yi_uncertainty status_flag" ;
        conc_4yi:coverage_content_type = "modelResult" ;
        conc_4yi:standard_name = "sea_ice_area_fraction" ;
        conc_4yi:scale_factor = 0.001f ;
        conc_4yi:add_offset = 0.f ;
        conc_4yi:grid_mapping = "Lambert_Azimuthal_Equal_Area" ;
short conc_5yi(time, y, x) ;
        conc_5yi:_FillValue = -32768s ;
        conc_5yi:long_name = "Concentration of Fifth Year Sea Ice" ;
        conc_5yi:name = "conc_5yi" ;
        conc_5yi:units = "1" ;
        conc_5yi:valid_min = 0s ;
        conc_5yi:valid_max = 1100s ;
        conc_5yi:ancillary_variables = "conc_5yi_uncertainty status_flag" ;
        conc_5yi:coverage_content_type = "modelResult" ;
        conc_5yi:standard_name = "sea_ice_area_fraction" ;
        conc_5yi:scale_factor = 0.001f ;
        conc_5yi:add_offset = 0.f ;
        conc_5yi:grid_mapping = "Lambert_Azimuthal_Equal_Area" ;
short conc_6yi(time, y, x) ;
        conc_6yi:_FillValue = -32768s ;
        conc_6yi:long_name = "Concentration of Sixth Year Sea Ice" ;
        conc_6yi:name = "conc_6yi" ;
        conc_6yi:units = "1" ;
        conc_6yi:valid_min = 0s ;
        conc_6yi:valid_max = 1100s ;
        conc_6yi:ancillary_variables = "conc_6yi_uncertainty status_flag" ;
        conc_6yi:coverage_content_type = "modelResult" ;
        conc_6yi:standard_name = "sea_ice_area_fraction" ;
        conc_6yi:scale_factor = 0.001f ;
        conc_6yi:add_offset = 0.f ;
        conc_6yi:grid_mapping = "Lambert_Azimuthal_Equal_Area" ;
short conc_7yi(time, y, x) ;
        conc_7yi:_FillValue = -32768s ;
        conc_7yi:long_name = "Concentration of Seventh Year Sea Ice" ;
        conc_7yi:name = "conc_7yi" ;
        conc_7yi:units = "1" ;
        conc_7yi:valid_min = 0s ;
        conc_7yi:valid_max = 1100s ;
        conc_7yi:ancillary_variables = "conc_7yi_uncertainty status_flag" ;
        conc_7yi:coverage_content_type = "modelResult" ;
```

```
            conc_7yi:standard_name = "sea_ice_area_fraction" ;
            conc_7yi:scale_factor = 0.001f ;
            conc_7yi:add_offset = 0.f ;
            conc_7yi:grid_mapping = "Lambert_Azimuthal_Equal_Area" ;

// global attributes:
            :title = "Arctic Sea Ice Age Climate Data Record v2.1" ;
            :summary = "This climate data record of sea ice age is obtained from coarse
resolution ice drift and conentration OSI SAF products. The processing chain features: 1)
Lagrangian advection of ice age fractions, 2) Weighted averaging of fractions." ;
            :iso_topic_category = "oceans,climatologyMeteorologyAtmosphere" ;
            :keywords = "GCMDSK:Earth Science > Cryosphere > Sea Ice > Sea Ice
Concentration, GCMDSK:Earth Science > Oceans > Sea Ice > Sea Ice Concentration,
GCMDSK:Earth Science > Climate Indicators > Cryospheric Indicators > Sea Ice
Concentration, GCMDSK:Earth Science > Cryosphere > Sea Ice > Sea Ice Motion,
GCMDSK:Earth Science > Oceans > Sea Ice > Sea Ice Motion, GCMDSK:Earth Science >
Climate Indicators > Cryospheric Indicators > Sea Ice Motion, GCMDLOC:Geographic Region
> Northern Hemisphere, GCMDLOC:Vertical Location > Sea Surface, GCMDPROV:
CONSORTIA/INSTITUTIONS > NERSC > Nansen Environmental and Remote Sensing
Center" ;
            :keywords_vocabulary = "GCMDSK:GCMD Science
Keywords:https://gcmd.earthdata.nasa.gov/kms/concepts/concept_scheme/sciencekeywords,
GCMDPROV:GCMD
Providers:https://gcmd.earthdata.nasa.gov/kms/concepts/concept_scheme/providers,
GCMDLOC:GCMD
Locations:https://gcmd.earthdata.nasa.gov/kms/concepts/concept_scheme/locations" ;
            :geospatial_lat_max = 90. ;
            :geospatial_lat_min = 17.61 ;
            :geospatial_lon_max = 180. ;
            :geospatial_lon_min = -180. ;
            :geospatial_vertical_min = 0. ;
            :geospatial_vertical_max = 0. ;
            :instrument = "Special Sensor Microwave - Imager (SSM/I), Special Sensor
Microwave - Imager (SSM/I), Special Sensor Microwave - Imager (SSM/I), Special Sensor
Microwave - Imager (SSM/I), Special Sensor Microwave - Imager (SSM/I), Special Sensor
Microwave - Imager (SSM/I), Special Sensor Microwave - Imager/Sounder (SSMIS), Special
Sensor Microwave - Imager/Sounder (SSMIS), Special Sensor Microwave - Imager/Sounder
(SSMIS), Advanced Microwave Scanning Radiometer for EOS (AMSR-E), Advanced
Microwave Scanning Radiometer - 2 (AMSR2)" ;
            string :platform = "Defense Meteorological Satellite Program - F08 (DMSP-F08),
Defense Meteorological Satellite Program - F10 (DMSP-F10), Defense Meteorological
Satellite Program - F11 (DMSP-F11), Defense Meteorological Satellite Program - F13 (DMSP-
F13), Defense Meteorological Satellite Program - F14 (DMSP-F14), Defense Meteorological
Satellite Program - F15 (DMSP-F15), Defense Meteorological Satellite Program - F16 (DMSP-
F16), Defense Meteorological Satellite Program - F17 (DMSP-F17), Defense Meteorological
Satellite Program - F18 (DMSP-F18), Earth Observation System - Aqua (Aqua), Global
Change Observation Mission for Water  (GCOM-W)" ;
            :platform_vocabulary = "WMO OSCAR Space:https://space.oscar.wmo.int/satellites"
;
            :instrument_vocabulary = "WMO OSCAR
Space:https://space.oscar.wmo.int/instruments" ;
            :source = "Daily Low Resolution Sea Ice Displacement from OSI SAF EUMETSAT
(OSI-405), \n Sea Ice Concentration Interim Climate Data Record Version 3 from the
EUMETSAT OSI SAF" ;
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:time_coverage_start = "2025-07-01T00:00:00Z" ;
:time_coverage_end = "2025-07-02T00:00:00Z" ;
:time_coverage_duration = "P1D" ;
:time_coverage_resolution = "P1D" ;
:project = "Thickness of Arctic sea ice Reconstructed by Data assimilation and artificial Intelligence Seamlessly (TARDIS)" ;
:institution = "Nansen Environmental and Remote Sensing Center" ;
:creator_name = "Anton Korosov" ;
:creator_type = "person" ;
:creator_url = "https://nersc.no/en/ansatt/anton-korosov/" ;
:creator_email = "anton.korosov@nersc.no" ;
:creator_institution = "Nansen Environmental and Remote Sensing Center (NERSC)" ;
:contact = "Anton Korosov, anton.korosov@nersc.no" ;
:license = "CC-BY-4.0" ;
:references = "https://doi.org/10.5194/essd-2025-477 (Scientific publication:A Climate Data Record of Sea Ice Age Using Lagrangian Advection of a Triangular Mesh)" ;
:date_created = "2026-02-12T00:00:00Z" ;
:cdm_data_type = "Grid" ;
:spatial_resolution = "65.0 km grid spacing" ;
:algorithm = "lagrangian_sea_ice_age_v2p1" ;
:geospatial_bounds_crs = "EPSG:6931" ;
:contributor_name = "Leo Edel, Laurent Bertino" ;
:contributor_email = "leo.edel@nersc.no, laurent.bertino@nersc.no" ;
:contributor_institution = "NERSC, NERSC" ;
:contributor_role = "assistant, project leader" ;
:Conventions = "CF-1.7, ACDD-1.3" ;
:standard_name_vocabulary = "CF Standard Name Table (Version 78, 21 September 2021)" ;
:product_name = "nersc_arctic_sea_ice_age_climate_data_record" ;
:product_id = "SEAICE_ARC_PHY_SIAGE_L4_MY_011_025" ;
:product_version = "2.1.2" ;
:product_status = "released" ;
:history = "Created on 2026-02-12T00:00:00Z by NERSC Sea Ice Age processing script" ;
:acknowledgment = "Research Council of Norway (project \'TARDIS\', no. 325241) and the European Space Agency (project \'CCI SAGE\', no. 4000147560/25/I-LR)" ;
:comment = "No comments" ;
:geospatial_bounds = "POLYGON((-5387500 -5387500, 5387500 -5387500, 5387500 5387500, -5387500 5387500, -5387500 -5387500))" ;
:processing_level = "Level 4" ;